Technical Report: AI-Powered Multi-Object Sports Tracking

Project Title: Multi-Object Detection and Persistent ID Tracking in Sports Footage

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1. Abstract

This project presents an end-to-end Computer Vision pipeline designed for the high-motion environment of sports analysis. The system integrates deep learning-based detection (YOLOv8) with advanced temporal tracking (BoT-SORT) to maintain subject identity and visualize movement trajectories. The final solution is delivered via an interactive Streamlit dashboard, optimized for both accuracy and user experience.

2. Problem Statement

Tracking athletes in sports footage presents several unique challenges:

- **Rapid Motion:** High-speed movement causes motion blur, affecting detection confidence.
- **Occlusion:** Players frequently overlap or move behind one another, leading to ID switches.
- **Camera Dynamics:** Panning, tilting, and zooming can confuse standard trackers that do not account for background motion.

3. Technical Methodology

3.1 Object Detection (YOLOv8m)

We utilized the **YOLOv8m (Medium)** model for its balance of inference speed and Mean Average Precision (mAP). It excels at detecting small-scale objects (players in the distance) and remains robust under varied lighting conditions.

3.2 Persistent Tracking (BoT-SORT)

Unlike standard trackers, **BoT-SORT** incorporates **Camera Motion Compensation (CMC)**. By using global motion estimation, the tracker can distinguish between the subject's actual movement and the camera's movement, ensuring IDs remain stable during pans and zooms.

3.3 Trajectory Mapping

A custom logic was implemented to store the center-point coordinates of each ID across frames. These coordinates are rendered as "tails" or trajectories, providing a visual history of a player's positioning and speed patterns over time.

4. System Architecture

The project follows a modular, Object-Oriented Programming (OOP) structure to ensure maintainability:

- **tracker.py**: Encapsulates the tracking engine and trajectory logic.
- **utils.py**: Handles video I/O, frame properties, and H.264 encoding for web compatibility.
- **app.py**: Implements the Streamlit-based Frontend with custom CSS injection for a professional "Dark Purple" dashboard aesthetic.

5. Implementation & Performance

- **Confidence Threshold:** Set at **0.3** to maintain detections during high-speed motion.

- **Inference Environment:** Developed and tested on **Antigravity.ai** utilizing Cloud GPU resources for high-speed frame processing.
- **Web Compatibility:** Implemented **AVC1 (H.264)** encoding to ensure processed output videos are playable directly in modern browsers without external players.

6. Results and Observations

The system successfully maintains ID persistence during complex player crossovers (occlusions). While CPU-based inference remains a bottleneck for real-time processing, the pipeline is fully optimized for GPU acceleration, where it can achieve 30+ FPS.

7. Conclusion & Future Scope

This project successfully demonstrates the application of SOTA (State-of-the-Art) AI models in a sports context. Future enhancements could include:

- **Action Recognition:** Classifying specific movements (e.g., jumping, sprinting).
- **Automated Speed Calculation:** Converting pixel distance to real-world meters for athletic performance metrics.
- **Multi-Camera Fusion:** Tracking players across different camera angles simultaneously.